

Scenario

I. Core Entities and Login Security

Every active entity in the system—Admin, Teacher, Student, and Parent—must authenticate using a unique username and password.

- Admin (First Name, Last Name): Holds complete system control and access to do anything (edit, delete, add users/data). They handle parent requests, message parents back, and can print a report for any specific student or teacher.
- Subject: Every Subject offered in the school has a unique name and a description.

II. Student, Grades, and Academic Structure

The structure for students and the tracking of their performance is strictly defined:

- Student (First Name, Last Name): Each student is assigned to a specific Class and belongs to a specific Academic Year. They are restricted to taking not more than seven subjects.
- Crucially, each student must have and maintain grades in each subject they are taking.
- They also maintain individual Attendance Records for their classes.
- Class: Each class is a physical location defined by a Class Number, is located on a specific Floor, and has a Capacity not more than 30 students.

III. Teacher and Parent Constraints

- Teacher (First Name, Last Name): Teaches only one subject but can teach up to three different classes of that subject per academic year. Their access is strictly limited to the subject and classes they teach.
- Teacher Actions: Teachers can publish a new announcement to their assigned classes and publish a new grade. When a grade is published, it triggers an instant notification to the student and is made available within the system. Teachers can also send a message to a specific student in their class.
- Parent (First Name, Last Name): A student's account can be connected to not more than two parents (father and mother). They cannot view other students' reports. They can initiate communication by messaging the Admin with problems or by requesting an official report for their child.

IV. Communication Rules

Communication is strictly governed:

- Student-to-Teacher Messaging: Students can send direct messages only to the teachers who teach them to ask questions. Students are explicitly prevented from messaging each other on the system.
- Parent-to-Admin Communication: Parents communicate directly with the Admin for report requests and problem resolution.